ON SYMMETRIC FUNCTIONS AND SEMINVARIANTS (continued)

[932, pp. 329–332 + lithographed Tables]

83. I do not wish in the present paper to go into the theory of covariants, but it is nevertheless proper to point out the connexion which exists between the covariant theory of derivation and the operators P and Q.

Consider a quantic $(a, b, c, \ldots, a' \mid x, y)^{\sigma'}$; any covariant hereof is $(A, B, C, \ldots \mid x, y)^{\sigma}$, where A is a seminvariant say of degree δ and weight $\varpi = \frac{1}{2}(\sigma'\delta - \mu)$, or $\mu = \sigma'\delta - 2\varpi$, reduced to zero by the operation $\Delta = a\partial_b + 2b\partial_c + \ldots + \sigma' a^{(\sigma'-1)}\partial_{a'}$: and if we write

$$\phi_\sigma = \sigma' b\partial_a + (\sigma - 1)c\partial_b + \ldots + \sigma a\partial_{a'},$$

then

$$B = \phi_\sigma A, \quad C = \tfrac{1}{2}\phi_\sigma B, \quad D = \tfrac{1}{3}\phi_\sigma C, \ \ldots$$

The derivative (f, F) is

$$\begin{aligned}(f, F) &= \partial_x f \cdot \partial_y F - \partial_y f \cdot \partial_x F \\ &= (a, b, \ldots \mid x, y)^{\sigma'-1} \cdot \sigma B x^{\sigma-1} + \ldots \\ &\quad - (b, c, \ldots \mid x, y)^{\sigma-1} \cdot \sigma' A x^{\sigma'-1} + \ldots\end{aligned}$$

that is, A being a seminvariant, we have $\sigma B - \mu b A$ a seminvariant, or say

$$(\phi_\sigma - \mu b)\, A = \text{sem.}, \quad \mu = \sigma'\delta - 2\varpi;$$

and similarly

$$(\phi_{\sigma'} - \mu' b)\, A = \text{sem.}, \quad \mu' = \sigma\delta - 2\varpi.$$

Hence

$$\{\sigma\phi_{\sigma'} - \sigma'\phi_\sigma - (\sigma\mu' - \sigma'\mu)\, b\}\, A, \text{ and } \{\sigma'\phi_{\sigma'} - \sigma\phi_\sigma - (\sigma'\mu' - \sigma\mu)\, b\}\, A$$

are each of them a seminvariant: but

$$\begin{aligned}\phi_\sigma &= \sigma\, b\partial_a + (\sigma - 1)\, c\partial_b + \ldots, \\ \phi_{\sigma'} &= \sigma'\, b\partial_a + (\sigma' - 1)\, c\partial_b + \ldots, \\ \phi_\sigma - \phi_{\sigma'} &= (\sigma - \sigma')\,(b\partial_a + c\partial_b + \ldots) = (\sigma - \sigma')\, P, \quad \mu - \mu' = (\sigma - \sigma')\, \delta;\end{aligned}$$

the first form, omitting the factor $\sigma - \sigma'$, is $= (P - \delta b)\, A$: similarly

$$\sigma'\phi_\sigma - \sigma\phi_{\sigma'} = (\sigma - \sigma')\,(c\partial_b + 2d\partial_c + \ldots) = (\sigma - \sigma')\, Q,$$

and

$$\sigma'\mu - \sigma\mu' = (\sigma - \sigma')\, 2\varpi,$$

and the second form is

$$(Q - 2\varpi b)\, A.$$

We thus see that the operators $P - \delta b$ and $Q - 2\varpi b$ upon a seminvariant A depend on the derivation of f upon a covariant which has A for its leading coefficient: the order of f is arbitrary, and we have thus two distinct forms.

84. As an illustration, consider the quantics

$$(1, b, c, d, e, f \mid x, y)^5,$$

and

$$(1, b, c, d, e, f, g \mid x, y)^6 :$$

each of these has a covariant the leading coefficient of which is

$$A = f - 5be + 2cd + 8b^2d - 6bc^2,$$

viz. these are

$$\left(\begin{array}{r} f+1 \\ be-5 \\ cd+2 \\ b^2d+8 \\ bc^2-6 \\ \hline \pm 11 \end{array}\right)(x,y)^5, \text{ and } \left(\begin{array}{r} f+1 \\ bf+5 \\ ce-16 \\ d^2+6 \\ b^2e-9 \\ bcd+38 \\ c^3-24 \\ \hline \pm 49 \end{array}\right)(x,y)^6;$$

and we find without difficulty

		$(g\infty d^2)$	$(ce\infty c^3)$	$(d^2\infty b^2c^2)$
(f_1, F_1)		-16	-10	
(f_2, F_2)	$=1$	-34	-16	
$(P-3b)\,A$	$=1$	-18	-6	
$(Q-10b)\,A$	$=5$	-74	-20	

and thence

$$(P-3b)\,A = (f_1, F_1) - (f_2, F_2),$$

$$(Q-10b)\,A = 5\,(f_1, F_1) - 6\,(f_2, F_2),$$

viz. we thus have $P-3b$, and $Q-10b$ upon $f\infty bc^2$ each given as a linear function of the derivatives (f_1, F_1) and (f_2, F_2), where f_1, f_2 are the quintic and the sextic function, and F_1, F_2 are like covariants of these functions respectively.

[Two lithographed Tables of seminvariants, occupying the whole of printed p. 331 (PDF p. 353), namely the Tables for $w = 11$ and $w = 12$. Each Table is a large square grid: the rows are labelled along the left edge by the monomial leading-terms (in alphabetical order, starting at b^k at the top and ending at c^k or d^k at the bottom), and the columns are labelled along the bottom edge by the corresponding power-enders (in alphabetical order, starting at b^k at the left and ending at the highest power-ender on the right). In each cell of the grid a small integer (or blank) gives the coefficient of the power-ender in the seminvariant whose leading term is the row label; the leading diagonal carries 1's. These two Tables continue the sequence of seminvariant Tables for $w = 1, 2, \ldots, 10$ already published in the earlier volumes of the Collected Papers.] [The Tables for $w = 13, 14, 15, 16$ are given on the accompanying lithographed sheet.]

[Lithographed Table for $w = 13$, occupying one full sheet (PDF p. 355), bearing the imprint "Cayley's Papers XIII. To follow p. 332." Same grid construction as the Tables for $w = 11$ and $w = 12$ above, but enlarged to accommodate the larger number of leading terms and power-enders of weight 13.]

[Lithographed Table for $w = 14$ (PDF p. 356), one full sheet, of the same grid construction as the Table for $w = 13$ but again enlarged for weight 14.]

[Lithographed Table for $w = 15$ (PDF p. 357), one full sheet, of the same grid construction, enlarged for weight 15.]

[Lithographed Table for $w = 16$ (PDF p. 358), one full sheet, of the same grid construction, enlarged for weight 16. These four lithographed sheets together extend the published series of seminvariant Tables to the weight 16.]

933.

TABLES OF PURE RECIPROCANTS TO THE WEIGHT 8.

[From the *American Journal of Mathematics*, t. XV. (1893), pp. 75–77.]

IN the tabulation of Pure Reciprocants it is convenient to write $a = 1$; we thus have for all the reciprocants of a given weight a single column of literal terms which (as in the Seminvariant Tables) I arrange in alphabetical order AO, and the several reciprocants have then each of them its own column of numerical coefficients: the form of the table is thus similar to that of the seminvariant table, the only difference being that for reciprocants the final terms are not in general power-enders: as in the seminvariant table, the columns of the table are arranged *inter se* with their final terms in AO. As remarked in my paper, "Corrected Seminvariant Tables for the Weights 11 and 12," *Amer. Math. Journ.*, t. XIV. (1892), pp. 195–200, [926], it is not in every case the top term of a column which should be regarded as the initial term; but to the extent 8, to which the reciprocant tables are here carried, this remark has no application.

I recall that the notation is the modified one employed by Halphen, and by Sylvester* in his 12th and subsequent lectures, viz. $a, b, c, d, \ldots$ denote

$$\frac{dy}{dx},\quad \frac{1}{2}\frac{d^2y}{dx^2},\quad \frac{1}{6}\frac{d^3y}{dx^3},\quad \frac{1}{24}\frac{d^4y}{dx^4},\ \ldots$$

respectively. As already noticed, a is put $= 1$, but it is to be in the several terms restored in the proper powers so as to obtain for the reciprocant a homogeneous expression of a degree equal to the original degree of the final term; thus $d - 3bc + 2b^3$ is to be read as standing for $a^2d - 3abc + 2b^3$.

The ultimate verification of the expression for a pure reciprocant consists (as is known) in its annihilation by the operator

$$V = 2a^2\partial_b + 5ab\partial_c + (6ac + 3b^2)\,\partial_d + (7ad + 7bc)\,\partial_e + (8ae + 8bd + 4c^2)\,\partial_f + \&\text{c.},$$

or, say

$$V = 2\partial_b + 5b\partial_c + (6c + 3b^2)\,\partial_d + (7d + 7bc)\,\partial_e + (8e + 8bd + 4c^2)\,\partial_f + \&\text{c.};$$

* *American Journal of Mathematics*, t. IX. (1887), p. 7.

thus for the reciprocant $50e - 175bd + 28c^2 + 105b^2c$, the result obtained is

$$2\,(-175\,d + 210\,bc) + 5b\,(-175\,c + 105\,b^2) + (6c + 3b^2)\,(-175\,b) + (7d + 7bc)\,(50),$$

or, collecting, this is

$$\begin{array}{r|rrrrr} d & -350 & +350 & & & \pm 350 \\ bc & +420 & +280 & -1050 & +350 & \pm 1050 \\ b^3 & & +525 & -525 & & \pm 525, \end{array}$$

$= 0$, as it should be.

The tables are

c	$+4$
b^2	-5
	± 4
	-5

d	$+1$
bc	-3
b^3	$+2$
	± 3

e	$+50$	
bd	-175	
c^2	$+28$	$+16$
b^2c	$+105$	-40
b^4		$+25$
	$+183$	$+41$
	-175	-40

f	$+10$	
be	-40	
cd	-12	$+1$
b^2d	$+63$	-5
bc^2	$+16$	-12
b^3c	-39	$+23$
b^5		-10
	± 91	± 27

g	$+14$			
bf	-63			
ce	-1350	$+800$		
d^2	$+1470$	-875		$+125$
b^2f	$+1782$	-1000		
bcd	-4158	$+2450$		-750
c^3	$+2130$	-1344	$+256$	$+64$
b^3e		$+500$		
b^2c^2		$+35$	$+165$	-240
b^4c			-300	$+500$
b^6				-125
	$+5576$	$+3250$	$+1018$	$+364$
	-5508	-3254		-365

i	$+420$						
bh	-2310						
cg	-32704	$+1176$					
df	$+57750$	-8085	$+20433$				
e^2	-20460	$+7040$	-21542	$+625$			
b^2g	$+45500$	-1470					
bcf	$+306$	-290	-150				
bde	-9090	$+520$	-16940	$+69062$	-4375		
c^2e	$+103740$	-27160	$+80248$	-49700	$+3200$	$+500$	
cd^2	-90900	$+38320$	$+26460$	-85554	$+55125$	-3500	-625
b^3f	$+69615$	-9555	$+40666$	-3000			
b^2ce	$+83538$	$+28098$	-106218	$+128625$	-8000		
b^2d^2	$+92820$	-52822	$+40866$	-156800	$+12740$	$+1024$	
bc^2d	-102102	$+21560$	-73304	$+191590$	-125200	$+4375$	-2025
c^4		-1344	$+21560$	$+84868$	$+125005$	$+9800$	-5750
b^4e		$+35$	$+1176$	-102165	$+5376$	-2560	
b^3cd			$+125000$	-78750	$+15$	$+1500$	$+256$
b^2c^3			-378	$+183750$	$+175$	$+6580$	-1280
b^5d				$+183768$			-2000
b^4c^2							$+2400$
b^6c							$+625$
b^8							-1280
	$+383768$	$+116037$	$+403375$	$+452993$	$+29130$	$+8774$	$+3281$
	-384803	-116032	-403077	-453040	-29126	-8750	-3280

[The numerical content of the larger tables is transcribed from the surviving scan as faithfully as possible; in a small number of cells the printed digits are partly broken or overprinted in the original lithography, and where the reading was uncertain the value most consistent with the column totals printed at the foot of each table has been retained.]

I remark that in the last of these tables the first column, say $i\infty bc^2d$, which ends in b^3cd, is a more simple form than Sylvester's P_8, $= i\infty c^4$, (*Amer. Math. Journ.*, t. IX. p. 35), which ends in c^4; P_8 is in fact a linear combination, first col. $+6$ second col., of the first and second columns of the table: the second column, say $eg\infty c^4$, is Sylvester's (a^2cg), t. IX. p. 124.

934.

NOTE ON THE SO-CALLED QUOTIENT G/H IN THE THEORY OF GROUPS.

[From the *American Journal of Mathematics*, t. XV. (1893), pp. 387, 388.]

THE notion (see Hölder, "Zur Reduction der algebraischen Gleichungen," *Math. Ann.*, t. XXXIV. (1887), §4, p. 31) is a very important one, and it is extensively made use of in Mr Young's paper, "On the Determination of Groups whose Order is the Power of a Prime," *American Journal of Mathematics*, t. XV. (1893), pp. 124–178; but it seems to me that the meaning is explained with hardly sufficient clearness, and that a more suitable algorithm might be adopted, viz. instead of $G = \Gamma_1,\ QG_1$, or $QG_1,\ \Gamma_1$, I would rather write $G = \Gamma_1{\cdot}QG_1$ or $QG_1{\cdot}\Gamma_1$.

We are concerned with a group G containing as part of itself a group Γ_1, such that each element of Γ_1 is commutative with each element of G. This being so, we may write

$$G = QG_1 \cdot \Gamma_1,$$

where QG_1 is not a group but a mere array of elements, viz. if $\Gamma_1 = (1, A_2, \ldots, A_s)$ and $QG_1 = (1, B_2, \ldots, B_t)$, then the formula is

$$G = (1, B_2, \ldots, B_t)\,(1, A_2, \ldots, A_s),$$

where it is to be noticed that the elements B are not determinate; in fact, if A_θ be any element of Γ_1, we may, in place of an element B, write BA_θ, for

$$B(1, A_2, \ldots, A_s) \quad \text{and} \quad BA_\theta\,(1, A_2, \ldots, A_s)$$

are, in different orders, the same elements of G.

But, G being a group, the product of any two elements of G is an element of G; viz. we thus have in general

$$B_iA_i \cdot B_jA_j = B_kA_k,$$

that is,

$$B_iB_j = B_kA_kA_i^{-1}A_j^{-1} \quad (i, j,\ \text{unequal or equal}),$$

where the B_k is a determinate element of the series $1, B_2, \ldots, B_t$, depending only on the elements B_i and B_j into the product of which it enters; and it is in nowise affected by the before-mentioned indeterminateness of the elements B: say B_i, B_j being any two elements of the series $1, B_2, \ldots, B_t$, we have the last preceding equation wherein B_k is a determinate element of the same series.

We may imagine a set of elements $1, B_2, \ldots, B_t$ for which, B_i, B_j being any two of them and B_k a third element determined as above, we have always $B_iB_j = B_k$, that is, these elements $1, B_2, \ldots, B_t$ now form a group, say the group G_1; the original elements $1, B_2, \ldots, B_t$ (which are subject to a different law of combination $B_iB_j = B_kA_kA_i^{-1}A_j^{-1}$ and do not form a group) are regarded as a mere array connected with this group, and so represented as above by QG_1; and the relation of the original group G to the group Γ_1 (consisting of elements commutative with those of G) and to the new group G_1 is expressed as above by the equation

$$G = \Gamma_1{\cdot}QG_1, = QG_1{\cdot}\Gamma_1.$$

Cambridge, 2 June, 1893.

935.

SUR LA FONCTION MODULAIRE $\chi\omega$.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. CXVI. (Janvier–Juin, 1893), pp. 1339–1343.]

SELON les notations usitées, on a

$$\chi\omega = \sqrt[4]{kk'} = 2^{\frac{1}{2}} q^{\frac{1}{8}} \cdot \frac{1+q}{1+q} \cdot \frac{1+q^3}{1+q^3} \cdot \frac{1+q^5}{1+q^5} \cdots$$

ou, d'après une transformation trouvée par M. Glaisher, on a

$$\frac{1}{1+q} \cdot \frac{3q^2}{1+q^3} \cdot \frac{5q^4}{1+q^5} \cdots = \frac{(1+q)^2}{(1-q^2)^2} \cdot \frac{q\,(1+q^3)^2}{(1-q^4)^2} \cdot \frac{q^2\,(1+q^5)^2}{(1-q^6)^2} \cdots$$

et de là, en intégrant,

$$\log\,(1+q)\cdot(1+q^3)\cdot(1+q^5)\cdots = \tfrac{1}{2}\frac{q}{1-q^2} - \tfrac{1}{4}\frac{q^2}{1-q^4} + \tfrac{1}{6}\frac{q^3}{1-q^6} - \cdots;$$

donc

$$\chi\omega = 2^{\frac{1}{2}} q^{\frac{1}{8}} \exp\left(\frac{q}{1-q^2} - \tfrac{1}{2}\frac{q^2}{1-q^4} + \tfrac{1}{3}\frac{q^3}{1-q^6} - \cdots\right),$$

ou, en écrivant $q = e^{i\pi\omega}$,

$$\begin{aligned}\chi\omega &= 2^{\frac{1}{2}} \exp\left[\frac{i\pi\omega}{24} + \frac{1}{2i}\left(\frac{1}{\sin\pi\omega} - \frac{1}{2\sin 2\pi\omega} + \frac{1}{3\sin 3\pi\omega} - \cdots\right)\right] \\ &= 2^{\frac{1}{2}} \exp\left[\frac{i\pi\omega}{24} - \frac{1}{2i}\sum \frac{\cos n\pi}{n\sin n\pi\omega}\right],\end{aligned}$$

$n = 1$ jusqu'à $n = \infty$, forme qui met en évidence l'équation

$$\chi\,(\omega+2) = i^{\frac{1}{4}}\chi\omega.$$

Mais nous avons

$$\frac{1}{n\sin n\pi\omega} = \frac{1}{n^2\pi\omega} + \frac{1}{\pi}\sum \frac{\cos m\pi}{n\,(n\omega - m)},$$

$m = 1$ jusqu'à $m = +\infty$ et $m = -1$ jusqu'à $m = -\infty$.

On a, dans les parenthèses, d'abord les termes

$$\frac{1}{2i}\frac{1}{\pi\omega}\left(\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \cdots\right) = \frac{1}{2i}\frac{1}{\pi\omega}\cdot\frac{\pi^2}{12} = -\frac{\pi i}{24\omega},$$

et l'expression de $\chi\omega$ devient ainsi

$$\chi\omega = 2^{\frac{1}{2}} \exp\left[\frac{i\pi\omega}{24} - \frac{i\pi}{24\omega} + \frac{i}{2\pi}\sum_n\sum_m \frac{\cos n\pi\cos m\pi}{n\,(n\omega - m)}\right],$$

ou, ce qui est plus commode,

$$\chi\omega = 2^{\frac{1}{2}} \exp\left[\frac{i\pi\omega}{24} - \frac{i\pi}{24\omega} + \frac{i}{2\pi}(S_1 - S_2 - S_3 + S_4)\,\frac{1}{n\,(n\omega - m)}\right],$$

où les signes sommatoires $S = \Sigma_n\Sigma_m$ se rapportent: S_1 aux valeurs impaires de m et n; S_2 aux valeurs impaires de m et paires de n; S_3 aux valeurs paires de m et impaires de n; S_4 aux valeurs paires de m et n. En omettant l'expression $\dfrac{1}{n\,(n\omega - m)}$ du terme sommé, on peut considérer S_1, S_2, S_3, S_4 comme dénotant les quatre sommes dont il s'agit.

Considérons d'abord S_4; en supposant que m', n' soient l'un ou l'autre ou tous les deux impairs, on peut donner à m, n les valeurs $2m', 2n'$; $4m', 4n'$; $8m', 8n'$; ...; on obtient

$$S_4 = (S_1 + S_2 + S_3)\left(\tfrac{1}{2^2} + \tfrac{1}{4^2} + \tfrac{1}{8^2} + \cdots\right) = \tfrac{1}{3}\,(S_1 + S_2 + S_3),$$

et le terme $S_1 - S_2 - S_3 + S_4$ dans l'expression de $\chi\omega$ devient ainsi

$$= \tfrac{4}{3}\,S_1 - \tfrac{2}{3}\,S_2 - \tfrac{2}{3}\,S_3.$$

Dans S_1, S_2, ou S_3, en supposant que m', n' dénotent des nombres relativement premiers, on peut donner à m, n les valeurs m', n'; $3m', 3n'$; $5m', 5n'$; ...: on obtient ainsi, pour chacune de ces sommes,

$$S = S\left(\tfrac{1}{1^2} + \tfrac{1}{3^2} + \tfrac{1}{5^2} + \cdots\right) = \tfrac{\pi^2}{8}\,S',$$

où, au côté droit, m et n sont des nombres relativement premiers; le terme devient ainsi

$$\frac{\pi^2}{6}\,(\tfrac{4}{3}S_1' - \tfrac{2}{3}S_2' - \tfrac{2}{3}S_3'),$$

et nous avons

$$\chi\omega = 2^{\frac{1}{2}}\exp\left[\frac{i\pi\omega}{24} - \frac{i\pi}{24\omega} + \frac{i\pi}{12}\,(S_1' - \tfrac{1}{2}S_2' - \tfrac{1}{2}S_3')\,\frac{1}{n\,(n\omega - m)}\right].$$

Je m'arrête pour remarquer que cette expression s'accorde parfaitement avec des résultats trouvés par M. Dedekind (Œuvres de Riemann, Leipzig, 1876, p. 447). En effet, en donnant à ω la valeur $\omega = \dfrac{m}{n} + i\alpha$, α une valeur positive très petite, M. Dedekind trouve les valeurs de $\log k$ et $\log k'$; en ajoutant ces valeurs et en ne faisant attention qu'aux termes qui deviennent infinis pour $\alpha = 0$, on a

$$\log kk' = 12\log\chi\omega = 24A,$$

m et n tous les deux impairs;

$$\log kk' = 12\log\chi\omega = -12A,$$

m et n l'un impair, l'autre pair.

Ici $A = \dfrac{\pi i}{24n\,(n\omega - m)}$, et les valeurs de $\log\chi\omega$ sont ainsi

$$\frac{i\pi}{12}\frac{1}{n\,(n\omega - m)} \quad \text{et} \quad -\frac{i\pi}{24}\frac{1}{n\,(n\omega - m)}$$

respectivement.

Dans l'expression de $\chi\omega$, il convient de réunir les termes qui correspondent aux valeurs $-m$, $+m$, c'est-à-dire au lieu de $\dfrac{1}{n\,(n\omega - m)}$, on doit écrire

$$\frac{1}{n}\left(\frac{1}{n\omega - m} + \frac{1}{n\omega + m}\right) = \frac{2\omega}{n^2\omega^2 - m^2};$$

on a ainsi finalement

$$\chi\omega = 2^{\frac{1}{2}}\exp\left[\frac{i\pi\omega}{24} - \frac{i\pi}{24\omega} + \frac{i\pi}{6}\,(S_1' - \tfrac{1}{2}S_2' - \tfrac{1}{2}S_3')\,\frac{\omega}{n^2\omega^2 - m^2}\right],$$

où je rappelle que m et n sont des nombres positifs relativement premiers, et que les signes sommatoires S_1, S_2, S_3 se rapportent, S_1 aux valeurs impaires de m et n, S_2 aux valeurs impaires de m et paires de n, S_3 aux valeurs paires de m et impaires de n.

Écrivant $-\dfrac{1}{\omega}$ au lieu de ω, le terme $\dfrac{\omega}{n^2\omega^2 - m^2}$ auquel se rapportent les sommations se change en $\dfrac{\omega}{n^2 - m^2\omega^2}$; on peut échanger les lettres m et n, et l'expression de $\chi\left(-\dfrac{1}{\omega}\right)$ devient ainsi identiquement celle de $\chi\omega$, c'est-à-dire la forme met en évidence la relation

$$\chi\left(-\frac{1}{\omega}\right) = \chi\omega.$$

Il y a cependant, dans cette analyse et par rapport à l'échange des lettres m et n, une difficulté qu'il convient d'écarter. Dans la formule

$$\frac{1}{\sin x} = \frac{1}{x} + \sum \frac{\cos m\pi}{x - m\pi} \quad (m = 1 \text{ jusqu'à } +\infty \text{ et } -1 \text{ jusqu'à } -\infty),$$

qui donne lieu à

$$\frac{1}{n \sin^2 n\pi\omega} = \frac{1}{n^2\pi^2\omega} + \frac{1}{\pi} \sum \frac{\cos m\pi}{n\,(n\omega - m)},$$

il est nécessaire que la variable x ait une valeur finie ou au moins infiniment petite par rapport aux valeurs extrêmes de m; ainsi, en écrivant pour x la valeur $n\pi\omega$, les valeurs extrêmes de n doivent être infiniment petites par rapport à celles de m, et la somme que nous avons dénotée simplement par $\sum_n \sum_m \dfrac{1}{n\,(n\omega - m)}$ signifie réellement

$$\sum_n^\nu \sum_m^\mu \frac{1}{n\,(n\omega - m)},$$

savoir les limites pour n sont $1, \nu$ et pour m ces limites sont $-1, -\mu$ et $+1, +\mu$ où μ et ν sont des nombres infiniment grands, mais $\dfrac{\mu}{\nu} = \infty$, ou, ce qui est la même chose, la somme est $\sum_n^\nu \sum_m^{\nu^2} \dfrac{1}{n\,(n\omega - m)}$, où ν est un très grand nombre qui devient enfin $= \infty$. En réunissant les termes pour m et $-m$, la somme à considérer est $\sum_n^\nu \sum_m^{\nu^2} \dfrac{1}{n^2\omega^2 - m^2}$, et il s'agit de faire voir qu'il est permis de substituer pour cela la somme

$$\sum_n^\nu \sum_m^\nu \frac{1}{n^2\omega^2 - m^2} \quad (\nu = \infty).$$

La différence des deux expressions est une somme double: $n = 1$ jusqu'à $n = \nu$ et $m = \nu + 1$ jusqu'à $m = \infty$, savoir cette somme est égale

$$\begin{aligned}
\cos\nu\pi &\left[+\frac{1}{\omega^2 - (\nu+1)^2} + \frac{1}{\omega^2 - (\nu+2)^2} + \frac{1}{\omega^2 - (\nu+3)^2} - \cdots\right] \\
&\quad - \left[\frac{1}{4\omega^2 - (\nu+1)^2} + \frac{1}{4\omega^2 - (\nu+2)^2} + \frac{1}{4\omega^2 - (\nu+3)^2} - \cdots\right] \\
\cos\nu\pi &\left[+\frac{1}{9\omega^2 - (\nu+1)^2} + \frac{1}{9\omega^2 - (\nu+2)^2} + \frac{1}{9\omega^2 - (\nu+3)^2} - \cdots\right] \\
&\quad - \cos\nu\pi \left[\frac{1}{\nu^2\omega^2 - (\nu+1)^2} + \frac{1}{\nu^2\omega^2 - (\nu+2)^2} + \frac{1}{\nu^2\omega^2 - (\nu+3)^2} - \cdots\right],
\end{aligned}$$

laquelle somme (sauf pour une valeur imaginaire $\omega = -\frac{1}{2} + i\alpha$, α positif) devient aussi petite que l'on veut en donnant à ν une valeur suffisamment grande; c'est-à-dire qu'on peut négliger cette différence et ainsi considérer la somme $\sum_n^\nu \sum_m^\nu \dfrac{1}{n\,(n\omega - m)}$, dont je me suis servi dans l'investigation comme ayant pour n et pour les valeurs positives ou négatives de m les mêmes limites $1, \nu$ $(\nu = \infty)$.

La forme trouvée pour $\chi\omega$ met en évidence que $\omega = 0$, $\omega = \infty$, et $\omega = \pm\dfrac{m}{n}$ sont des valeurs essentiellement singulières pour la fonction.

936.

NOTE ON UNIFORM CONVERGENCE.

[From the *Proceedings of the Royal Society of Edinburgh*, vol. XIX. (1893), pp. 203–207. Read December 5, 1892.]

It appears to me that the form in which the definition or condition of uniform convergence is usually stated, is (to say the least) not easily intelligible. I call to mind the general notion: We may have a series, to fix the ideas, say of positive terms

$$(0)_x + (1)_x + (2)_x + \ldots + (n)_x + \ldots,$$

the successive terms whereof are continuous functions of x, for all values of x from some value less than a up to and inclusive of a (or from some value greater than a down to and inclusive of a): and the series may be convergent for all such values of x, the sum of the series ϕx is thus a determinate function ϕx of x; but ϕx is not of necessity a continuous function; if it be so, then the series is said to be uniformly convergent; if not, and there is for the value $x = a$ a breach of continuity in the function ϕx, then there is for this value $x = a$ a breach of uniform convergence in the series.

Thus if the limits are say from 0 up to the critical value 1, then in the geometrical series $1+x+x^2+\ldots$, the successive terms are each of them continuous up to and inclusive of the limit 1, but the series is only convergent up to and exclusive of this limit, viz. for $x = 1$ we have the divergent series $1+1+1+\ldots$, and this is not an instance; but taking, instead, the geometrical series

$$(1-x) + (1-x)\,x + (1-x)\,x^2 + \ldots,$$

here the terms are each of them continuous up to and inclusive of the limit 1, and the series is also convergent up to and inclusive of this limit; in fact, at the limit $x = 1$,

the series is $0+0+0+\ldots$ We have here an instance, and there is in fact a discontinuity in the sum, viz. $x < 1$ the sum is

$$(1-x)\,(1+x+x^2+\ldots) = (1-x)\cdot\frac{1}{1-x} = 1,$$

whereas for the limiting value 1, the sum is $0+0+0+\ldots, = 0$. The series is thus uniformly convergent up to and exclusive of the value $x = 1$, but for this value there is a breach of uniform convergence.

I remark that Du Bois-Reymond in his paper, "Notiz über einen Cauchy'schen Satz, die Stetigkeit von Summen unendlicher Reihen betreffend," *Math. Ann.*, t. IV. (1871), pp. 135–137, shows that, when certain conditions are satisfied, the sum ϕx is a continuous function of x, but he does not use the term "uniform convergence," nor give any actual definition thereof.

M. Jordan, in his "Cours d'Analyse de l'École Polytechnique," t. I. (Paris, 1882), considers p. 116 the series $s = u_1 + u_2 + u_3 + \ldots$, the terms of which are functions of a variable z, and after remarking that such a series is convergent for the values of z included within a certain interval, if for each of these values and for every value of the infinitely small quantity ϵ we can assign a value of n such that for every value of p,

$$\mathrm{Mod}\,(u_{n+1} + u_{n+2} + \ldots + u_{n+p}) < \mathrm{Mod}\,\epsilon,$$

ϵ being as small as we please, proceeds:—

"Le nombre des termes qu'il est nécessaire de prendre dans la série pour arriver à ce résultat sera en général une fonction de z et de ϵ. Néanmoins on pourra très habituellement déterminer un nombre n fonction de ϵ seulement telle que la condition soit satisfaite pour toute valeur de z comprise dans l'intervalle considéré. On dira dans ce cas que la série s est uniformément convergente dans cet intervalle."

And similarly, Professor Chrystal in his Algebra, Part II. (Edinburgh, 1889), after considering, p. 130, the series

$$\frac{x}{x+1\cdot 2x+1} + \frac{x}{2x+1\cdot 3x+1} + \ldots + \frac{x}{(n-1)\,x+1\cdot nx+1} + \cdots$$

for which the critical value is $x = 0$, and in which when $x \neq 0$ the residue R_n of the series or sum of the $(n+1)$th and following terms is $= \dfrac{1}{nx+1}$, proceeds as follows:—

Now when x has any given value, we can by making n large enough make $\dfrac{1}{nx+1}$ smaller than any given positive quantity a. But on the other hand, the smaller x is the larger must we take n in order that

$\dfrac{1}{nx+1}$ may fall under a; and in general when x is variable there is no finite upper limit for n independent of x, say ν, such that if $n > \nu$ then $R_n < a$. When the residue has this peculiarity the series is said to be *non-uniformly convergent*; and if for a particular value of x, such as $x = 0$ in the

present example, the number of terms required to secure a given degree of approximation to the limit is infinite, the series is said to *converge infinitely slowly*.

And he thereupon gives the formal definition: If for values of x within a given region in Argand's diagram we can for every value of a, however small Mod. a, assign for n an upper limit ν INDEPENDENT of x, such that, when $n > \nu$, Mod. $R_n <$ Mod. a, then the series $\sum f(n,x)$ is said to be UNIFORMLY CONVERGENT within the region in question.

The two forms of definition (Jordan and Chrystal) appear to me equivalent, and it seems to me that construing the definition strictly, and applying it to the above instance

$$(1-x) + (1-x)\,x + (1-x)\,x^2 + \ldots,$$

the definition does not in either case indicate a breach of uniform convergency at $x = 1$, viz. the definition shows uniform convergency from $x = 0$ to $x = 1-\epsilon$, ϵ being a positive quantity however small, or (as I have before expressed this) uniform convergency up to and exclusive of the limit 1; and further, it shows uniform convergency at the limit 1. For at this limit, the series of terms is $0+0+0+\ldots$, the residue or sum of the $(n+1)$th and subsequent terms is thus also $0+0+0+\ldots$, and we get the value of this residue, not approximately, but exactly, by taking a single term of the series. Jordan and Chrystal calculate, each of them, the residue from the general expression thereof by writing therein for x or z the critical value; and then, comparing the value thus obtained with the values obtained for the $(n+1)$th and subsequent terms of the series on substituting therein for x or z the critical value, they seem to argue that the discrepancy between these two values indicates the breach of uniform convergency.

It may be said that the objection is a verbal one. But it seems to me that the whole notion of the residue (although very important as regards the general theory of convergence) is irrelevant to the present question of uniform convergency, and that a better method of treating the question is as follows:

Considering as before the series

$$(0)_x + (1)_x + (2)_x + \ldots + (n)_x + \ldots,$$

where the functions $(0)_x, (1)_x, (2)_x, \ldots$ are each of them continuous up to and inclusive of the limit $x = a$, and the series has thus a definite sum ϕx, this sum is *primâ facie* a continuous function of x, and what we have to explain is the manner in which it may come to be discontinuous. Suppose that it is continuous up to and exclusive of the limit $x = a$, but that there is a breach of discontinuity at this limit: write $x = a-\epsilon$, where ϵ is a positive quantity as small as we please, and consider the two equations

$$\phi a = (0)_a + (1)_a + (2)_a + \ldots,$$

$$\phi x = (0)_x + (1)_x + (2)_x + \ldots,$$

then we have

$$\phi a - \phi x = \epsilon \left\{ \frac{(0)_a - (0)_x}{a-x} + \frac{(1)_a - (1)_x}{a-x} + \ldots \right\}.$$

Hence if the sum of the series in { } is a finite magnitude M, not indefinitely large for an indefinitely small value of ϵ, we have $\phi a - \phi x = \epsilon M$, which is indefinitely small for ϵ indefinitely small, and there is no breach of continuity; the only way in which a breach of continuity can arise is by the series in { } having a value indefinitely large for ϵ indefinitely small, viz. if such a value is $\dfrac{N}{\epsilon}$, then $\phi a - \phi x = \epsilon \cdot \dfrac{N}{\epsilon} = N$, and as x changes from $a-\epsilon$ to a, the sum changes abruptly from $\phi(a-\epsilon)$ to $\phi(a-\epsilon) + N$.

The condition for a breach of uniform convergency for the value $x = a$, thus is that, writing $x = a-\epsilon$, ϵ a positive magnitude however small, the series

$$\frac{(0)_a - (0)_x}{a-x} + \frac{(1)_a - (1)_x}{a-x} + \ldots,$$

shall have a sum indefinitely large for ϵ indefinitely small, or say as before, a sum $= \dfrac{N}{\epsilon}$.

For the foregoing example, where the series is

$$(1-x) + x\,(1-x) + x^2\,(1-x) + \ldots,$$

the critical value is $a = 1$: we have here $(n)_x = x^n\,(1-x)$, and consequently

$$\frac{(n)_a - (n)_x}{a-x} = \frac{a^n - x^n}{a-x} - \frac{a^{n+1} - x^{n+1}}{a-x}$$

$$= (a^{n-1} + a^{n-2}x + \ldots + x^{n-1}) - (a^n + a^{n-1}x + \ldots + x^n)$$

$$= -x^n \quad \text{for } a = 1.$$

The series

$$\frac{(0)_a - (0)_x}{a-x} + \frac{(1)_a - (1)_x}{a-x} + \ldots$$

thus is

$$-1 - x - x^2 - \ldots = -\frac{1}{1-x} = -\frac{1}{\epsilon}$$

for $x = 1 - \epsilon$, viz. we have

$$\phi 1 - \phi(1-\epsilon) = \epsilon \cdot \left(-\frac{1}{\epsilon}\right) = -1;$$

which is right since by what precedes

$$\phi 1 = 0, \quad \phi(1-\epsilon) = 1.$$

937.

NOTE ON THE ORTHOTOMIC CURVE OF A SYSTEM OF LINES IN A PLANE.

[From the *Messenger of Mathematics*, vol. XXII. (1893), pp. 45, 46.]

CONSIDERING *in plano* a singly infinite system of lines, then to each point of the plane there corresponds a line (not in general a unique line), and we can therefore express in terms of the coordinates (x, y) of the point the cosine-inclinations α, β of the line to the axes. The differential equation of the orthotomic curve is then $\alpha\, dx + \beta\, dy = 0$, and it is a well-known and easily demonstrable theorem that $\alpha\, dx + \beta\, dy$ is a complete differential, say it is $= dV$; the integral equation of the orthotomic curve is therefore $V = \int (\alpha\, dx + \beta\, dy) =$ const., and we see further that V is a solution of the partial differential equation

$$\left(\frac{dV}{dx}\right)^2 + \left(\frac{dV}{dy}\right)^2 = 1.$$

If the lines are the normals of the ellipse $\dfrac{X^2}{a} + \dfrac{Y^2}{b} = 1$, then, writing the equation of the normal at the point X, Y in the form

$$\frac{a}{X}(x - X) = \frac{b}{Y}(y - Y) = \lambda,$$

suppose, we have

$$X = \frac{ax}{a+\lambda}, \quad Y = \frac{by}{b+\lambda};$$

and therefore

$$\frac{x^2}{(a+\lambda)^2} + \frac{y^2}{(b+\lambda)^2} = 1,$$

which last equation determines λ as a function of x, y. We have α, β proportional to $\dfrac{X}{a}, \dfrac{Y}{b}$; or say we have

$$\alpha = M\frac{x}{a+\lambda}, \quad \beta = M\frac{y}{b+\lambda},$$

whence

$$\frac{1}{M^2} = \frac{x^2}{(a+\lambda)^2} + \frac{y^2}{(b+\lambda)^2} = 1.$$

or, writing for convenience

$$\frac{x^2}{(a+\lambda)^2} + \frac{y^2}{(b+\lambda)^2} - \frac{k^2}{\lambda^2} = 0,$$

(viz. this equation defines k as a function of x, y and λ, that is, of x and y) we have

$$\alpha = \frac{\lambda x}{k(a+\lambda)}, \quad \beta = \frac{\lambda y}{k(b+\lambda)};$$

and we ought therefore to have

$$\frac{\lambda}{k}\left(\frac{x\, dx}{a+\lambda} + \frac{y\, dy}{b+\lambda}\right)$$

a complete differential, $= dV$.

This is easily verified, for from the assumed value

$$k = \lambda\left(\frac{x^2}{(a+\lambda)^2} + \frac{y^2}{(b+\lambda)^2} - 1\right)^{\frac{1}{2}},$$

we deduce

$$dk = 2\lambda\left(\frac{x\, dx}{a+\lambda} + \frac{y\, dy}{b+\lambda}\right) + d\lambda\left(\frac{x^2}{(a+\lambda)^2} + \frac{y^2}{(b+\lambda)^2} - 1\right) = 2\lambda\left(\frac{x\, dx}{a+\lambda} + \frac{y\, dy}{b+\lambda}\right),$$

and we have therefore

$$dV = \frac{\lambda}{k}\left(\frac{x\,dx}{a+\lambda} + \frac{y\,dy}{b+\lambda}\right) = \frac{1}{2k}\cdot 2k\,dk\cdot\frac{1}{k} = dk,$$

where k denotes a function of (x, y) defined as above; hence the equation $V =$ const. gives $k =$ const., or the equation of the orthotomic curve is given by the system of equations

$$\frac{x^2}{(a+\lambda)^2} + \frac{y^2}{(b+\lambda)^2} = 1, \quad \frac{x^2}{(a+\lambda)^2} + \frac{y^2}{(b+\lambda)^2} - \frac{k^2}{\lambda^2} = 0,$$

where k is a constant; these equations (eliminating λ) give, in fact, the equation of the parallel curve of the ellipse, and k denotes the normal distance of a point on the curve from the ellipse. I recall that the first equation may be replaced by

$$\frac{x^2}{a+\lambda} + \frac{y^2}{b+\lambda} = \lambda,$$

and since the derived equation hereof in regard to λ is the second equation, we have the equation of the parallel curve in the known form

$$\text{Disc.}\,\{(\lambda+k)(\lambda+a)(\lambda+b) - (b+\lambda)\,x^2 - (a+\lambda)\,y^2\} = 0.$$

I notice further that, considering k a function of x, y as above, we have

$$\left(\frac{dk}{dx}\right)^2 + \left(\frac{dk}{dy}\right)^2 = \frac{1}{4k^2}\left(\left(\frac{dk^2}{dx}\right)^2 + \left(\frac{dk^2}{dy}\right)^2\right),$$

that is,

$$\left(\frac{dV}{dx}\right)^2 + \left(\frac{dV}{dy}\right)^2 = 1,$$

as it should be.

938.

ON TWO CUBIC EQUATIONS.

[From the *Messenger of Mathematics*, vol. XXII. (1893), pp. 69–71.]

STARTING from the equations

$$2 + a = b^2, \quad 2 + b = c^2, \quad 2 + c = a^2,$$

then eliminating b, c, we find

$$(a^4 - 4a^2 + 2)^2 - (a + 2) = 0,$$

that is,

$$a^8 - 8a^6 + 20a^4 - 16a^2 - a + 2 = 0;$$

we satisfy the equations by $a = b = c$, and thence by

$$a^2 - a - 2 = (a - 2)(a + 1) = 0;$$

there remains a sextic equation breaking up into two cubic equations; the octic equation may in fact be written

$$(a - 2)(a + 1)(a^3 + a^2 - 2a - 1)(a^3 - 3a + 1) = 0,$$

and we have thus the two cubic equations

$$x^3 + x^2 - 2x - 1 = 0, \quad x^3 - 3x + 1 = 0,$$

for each of which the roots (a, b, c) taken in a proper order are such that $2 + a = b^2$, $2 + b = c^2$, $2 + c = a^2$.

It may be remarked that starting from $y^3 + y^2 - 2y - 1 = 0$, $y^2 = x + 2$, the first equation gives $(y^3 - 2y)^2 - (y^2 - 1)^2 = 0$, that is, $y^6 - 5y^4 + 6y^2 - 1 = 0$, whence

$$(x + 2)^3 - 5(x + 2)^2 + 6(x + 2) - 1 = 0,$$

that is,

$$x^3 + x^2 - 2x - 1 = 0.$$

And similarly, starting from $y^3 - 3y + 1 = 0$, $y^2 = x + 2$, the first equation gives $(y^3 - 3y)^2 - 1 = 0$, that is, $y^6 - 6y^4 + 9y^2 - 1 = 0$, whence

$$(x + 2)^3 - 6(x + 2)^2 + 9(x + 2) - 1 = 0,$$

that is,

$$x^3 - 3x + 1 = 0.$$

To find the roots of the equation $x^3 + x^2 - 2x - 1 = 0$, taking ω an imaginary cube root of unity, and writing $\alpha = \sqrt[3]{7(2 + 3\omega)}$, $\beta = \sqrt[3]{7(2 + 3\omega^2)}$, where observe that $2 + 3\omega$, $2 + 3\omega^2$ are imaginary factors of 7, viz.

$$7 = (2 + 3\omega)(2 + 3\omega^2),$$

and therefore also $\alpha^3 + \beta^3 = 7$, $\alpha\beta = 7$, then the roots of the equation are

$$3a = -1 + \alpha + \beta,$$

$$3b = -1 + \omega\alpha + \omega^2\beta,$$

$$3c = -1 + \omega^2\alpha + \omega\beta.$$

I verify herewith the equation $a^2 = 2 + c$, viz. this gives

$$(-1 + \alpha + \beta)^2 = 18 + 3(-1 + \omega^2\alpha + \omega\beta),$$

or writing herein $2\alpha\beta = 14$, this is

$$\alpha^2 - (1 + 3\omega^2)\alpha + \beta^2 - (1 + 3\omega)\beta = 0,$$

that is,

$$-\omega^2\,(\alpha\omega+\beta\omega^2)=0,$$

or finally

$$\omega^2\alpha\,(1-\omega\beta)=0,$$

satisfied in virtue of $\alpha\beta=7$.

For the second equation $x^3-3x+1=0$, ω denoting as before, the roots are

$$\begin{aligned} a&=\omega^{\frac{1}{3}}+\omega^{-\frac{1}{3}}, &\qquad \text{whence } a^2&=\omega^{\frac{2}{3}}+\omega^{-\frac{2}{3}}+2,=2+c,\\ b&=\omega^{\frac{4}{3}}+\omega^{-\frac{4}{3}}, & b^2&=\omega^{\frac{8}{3}}+\omega^{-\frac{8}{3}}+2,=2+a,\\ c&=\omega^{\frac{2}{3}}+\omega^{-\frac{2}{3}}, & c^2&=\omega^{\frac{4}{3}}+\omega^{-\frac{4}{3}}+2,=2+b.\end{aligned}$$

The equation $x^3-5x^2+6x-1=0$, which, writing therein $x+2$ for x, gives

$$x^3+x^2-2x-1=0,$$

is considered in Hermite's *Cours d'Analyse*, Paris 1873, p. 12, and this suggested to me the foregoing investigation.